

JEE MAIN REVISION MASTER: STRAIGHT LINES AND PAIR OF STRAIGHT LINES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 171 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Distance, Slopes & Forms of Straight Line	119 PYQs	Straight Lines
Very High	Pair of Straight Lines & Angle Bisectors	53 PYQs	Homogeneous Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Distance, Slopes & Forms of Straight Line	$d = \frac{ a x_1 + b y_1 + c }{\sqrt{a^2 + b^2}}$, $d_{\text{parallel}} = \frac{ c_1 - c_2 }{\sqrt{a^2 + b^2}}$, $\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$
Pair of Straight Lines & Angle Bisectors	$\tan \theta = \left \frac{2\sqrt{h^2 - ab}}{a + b} \right $, $\text{Perpendicular} \iff a + b = 0$, $\text{Coincident} \iff h^2 = ab$, $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Distance, Slopes & Forms of Straight Line (Very High Yield — 119 PYQs)

Core Principle: Linear equations, distance formulas, family of lines $L_1 + \lambda L_2 = 0$, and coordinates of reflections (images) in lines.

■ **Caution / JEE Pitfall:** For distance between parallel lines, ensure coefficients of x and y are MADE IDENTICAL before subtracting $c_1 - c_2$.

■ **Speed Trick:** Image of point (x_1, y_1) in line $ax + by + c = 0$ is given by $(x - x_1 / a) = (y - y_1 / b) = -2(ax_1 + by_1 + c) / (a^2 + b^2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If a straight line drawn through the point of intersection of the lines $4x+3y-1=0$ and $3x+4y-1=0$, meets the co-ordinate axes at the poin..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the mid points of the sides of a triangle ABC be $((5/2), 7)$, $((5/2), 3)$ and $(4, 5)$. If its incentre is $(\blacksquare, k)$, then \$..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Let $A(1, 2)$ and $C(-3, -6)$ be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x-y=14$. If ..." [Ans: B]

3.2 Pair of Straight Lines & Angle Bisectors (Very High Yield — 53 PYQs)

Core Principle: Joint equations of lines through origin, condition of perpendicularity ($a + b = 0$), and joint equation of angle bisectors.

■ **Caution / JEE Pitfall:** If $a + b = 0$, the lines represented by $ax^2 + 2hxy + by^2 = 0$ are mutually perpendicular regardless of value of h .

■ **Speed Trick:** General second degree equation represents pair of lines if $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "From the point $(-1, -1)$, two rays are sent making angles of 45° with the line $x+y=0$. These rays get reflected from the mirror ..." **[Ans: 7]**
- **[2026 Morning Shift] Q#2 (MCQ):** "Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x+2\sqrt{2}y=4$. If the co-ordinates of the ..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Let the line $L_1: x+3=0$ intersect the lines $L_2: x-y=0$ and $L_3: 3x+y=0$ at the points A and B, respectively. Let the bisector of the ..." **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let the mid points of the sides of a triangle ABC be $((5/2), 7)$, $((5/2), 3)$ and $(4, 5)$. If its incentre is (α, k) , then $3\alpha + k$ is equal to :

- (A) 11
- (B) 12
- (C) 13 **[Correct]**
- (D) 14

Correct Answer: C

Solution Analysis: Let the midpoints of the sides of triangle ABC be $D((5/2), 7)$, $E((5/2), 3)$, $F(4, 5)$. We have to find the incentre (α, k) of triangle ABC. Step 1: Use the fact about the medial triangle. The triangle formed by joining the midpoints of the sides of a triangle is called the medial triangle. If the midpoints are known, then the original triangle can be reconstructed using: $A = E + F - D$, $B = F + D - E$, $C = D + E - F$. Now calculate each vertex. Vertex A $A = ((5/2), 3) + (4, 5) - ((5/2), 7) = (4, 1)$. Vertex B $B = (4, 5) + ((5/2), 7) - ((5/2), 3) = (4, 9)$. Vertex C $C = ((5/2), 7) + ((5/2), 3) - (4, 5) = (5, 5)$.

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

From the point $(-1, -1)$, two rays are sent making angles of 45° with the line $x+y=0$. These rays get reflected from the mirror $x+2y=1$. If the equations of the reflected rays are $ax+by=9$ and $cx+dy=7$, $a, b, c, d \in \mathbb{Z}$, then the value of $ad+bc$ is ____.

Correct Answer: 7

Solution Analysis: Line $x+y=0$ has slope -1 . If a ray has slope m and makes an angle 45° with this line, then by the angle formula between two lines we use: $\tan 45^\circ = |(m+1)/(1-m)|$. Now $\tan 45^\circ = 1$, so we get: $\rightarrow \pm 1 = (m+1)/(1-m)$. Solving this gives two possible directions. One solution gives a horizontal line ($m=0$), and the other corresponds to a vertical line (perpendicular to the x -axis). $\rightarrow m=0$ & one line is perpendicular to x axis. Both rays start from $(-1, -1)$, so the two incident rays are: $L_1: x=-1$, $L_2: y=-1$. Next, each ray